

Java Intern Evaluation Exam

Section A: Short Programs (2 x 5 = 10 Marks)

1. Write a program to count character frequency in a string.
2. Write a program to reverse a string without using built-in methods.

Section B: Scenario-Based Coding (Any 2 × 20 = 40 Marks)

1. Library Management System

- Class `Book` with `title`, `author`, `ISBN`
- Class `Library` with `ArrayList<Book>`
- Add/search functionality, throw `BookNotFoundException`

2. E-commerce Payment Interface

- Interface `Payment` with `pay()` method
- Implement: `UPIPayment`, `CardPayment`, `CashOnDelivery`
- Demonstrate runtime polymorphism

3. Student Grades System

- Create `Student` class
- Store students in a List
- Sort by marks using Comparator

4. Bank ATM Simulation

- Class `BankAccount` with `deposit()`, `withdraw()`
- Encapsulation for balance
- Throw exceptions on negative deposit/insufficient balance